

Dentons - Ontario's Data Centre Playbook: New rules for a new era of electricity demand

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Ontario's Data Centre Playbook: New rules for a new era of electricity demand Data centres are the physical infrastructure behind the digital economy: every email, online transaction and artificial intelligence (AI) query depends on a warehouse-scale facility requiring a continuous, high-volume supply of electricity. Ontario is building its data centre framework as AI-driven demand reshapes electricity markets. Proposed data centre connection requests could exceed 10,000 MW, against a provincial peak demand of roughly 25,000 MW. Data centres are projected to account for about 13% of Ontario's new electricity demand by 2035. The IESO's 2026 Annual Planning Outlook forecasts data centre consumption rising as overall provincial demand grows by 65%. The pressure on the system is immediate, not hypothetical. Microsoft is constructing its YTO 11 facility in Vaughan while HIVE Digital Technologies has announced a 320 MW AI campus valued at CA\$3.5 billion. Ontario already hosts approximately 100 data centres and employs nearly 450,000 tech workers across 26,000 businesses, including more than 1,600 AI companies. The province is positioning itself as a North American data centre hub. Its response is organized around three pillars: (i) legislation, Bill 40; (ii) policy, the Data Centre Playbook; and (iii) technical standards, the Independent Electricity System Operator (IESO)'s requirements for large computational loads. Together, these measures determine who may connect, on what terms, at what cost and subject to what technical standards. This article explains the framework and its implications for developers, utilities, municipalities, investors and ratepayers. How Ontario is positioning itself for data centre growth Ontario's framework has developed over roughly two years against a broader investment backdrop. Ontario's approach is not simply to manage load growth; it is to prioritize data centre projects that align with broader economic-development and digital-sovereignty objectives. Each data centre project raises questions about energy and infrastructure, technology supply chains, privacy and data sovereignty, Indigenous engagement,

environmental constraints, community acceptance and foreign investment. Progressively, the following iterative developments have shaped Ontario's approach to data centres:

- In October 2024, the government released Ontario's Affordable Energy Future: The Pressing Case for More Power. The paper framed expanded electricity supply as part of a broader pro-growth agenda and signalled that energy planning would need to be coordinated across sectors.
- Bill 214, the Affordable Energy Act, 2024, followed shortly after. It requires Ontario to produce an Integrated Energy Plan on a five-year cycle, bringing electricity, natural gas and other fuels into a coordinated framework for the first time.
- On May 1, 2025, the IESO's Market Renewal Program introduced Locational Marginal Pricing, replacing the previous uniform pricing model with prices that reflect the cost of producing and consuming electricity at particular locations on the grid. For data centre developers, the change makes location a direct input into electricity cost and project feasibility. The framework then turned to access.
- On June 3, 2025, Ontario introduced Bill 40, the Protect Ontario by Securing Affordable Energy for Generations Act, 2025. The bill amended the Electricity Act, 1998 (the Electricity Act) and the Ontario Energy Board Act, 1998.
- Nine days later, Ontario published its first Integrated Energy Plan, Energy for Generations. The sequence was deliberate: supply planning, market design and a more selective connection regime were being developed in parallel.
- By July 2025, the IESO had identified large step loads, that is loads exceeding 20 MW that connect in large blocks over short periods, as a distinct planning issue. In September, Ontario published an ERO notice seeking input on data centre connection regulations.
- On December 11, 2025, Bill 40 received Royal Assent. In 2026, implementation became the focus.
- The IESO posted draft Technical Requirements for Large Computational Loads (Version 2.0) in May and held a public engagement in July. On August 13, 2026, Ontario launched the Data Centre Playbook. The related 30-day ERO consultation closed September 12, 2026. In September 2026, the IESO issued Version 3.0 of its Technical Requirements, extending coverage to distribution-connected projects. Bill 40: From open access to merit-based connection decisions Historically, Ontario's electricity framework generally favoured non-discriminatory access to the grid where technical, economic and reliability requirements could be satisfied. Bill 40 introduces an additional policy-based approval layer for certain large loads by amending, among other acts, the Electricity Act. New section 28.1 of the Electricity Act creates an exception for "specified load facilities," including data centres and any other facility exceeding a prescribed demand threshold. Transmitters and distributors may not connect or reconnect these facilities unless they meet requirements established by regulation. The result is a distinct approval track for the largest loads, not an automatic connection entitlement. Facilities that submitted a connection request before June 3, 2025 are exempt from the section 28.1 approval process, though they must still meet IESO technical requirements if their SIA or Notice of Conditional Approval is issued after June 30, 2026 as further detailed below. The regulation-making authority prescribed under the Electricity Act is deliberately broad. Connection requirements may address economic development, job creation and any other matter the Lieutenant Governor in Council considers necessary or advisable. The new regime contemplates a significant ministerial role in determining whether specified load facilities satisfy the applicable connection requirements established by regulation. A data centre application will therefore be evaluated not only on technical and financial grounds, but also on the economic, strategic and community benefits it offers Ontario. Primacy over the Ontario Energy Board Section 28.1 prevails over any order,

code or licence issued by the Ontario Energy Board in the event of a conflict. This override is significant because it gives the specified-load regime priority where a conflict arises between section 28.1 and OEB-issued instruments. As a practical matter, it shifts greater decision-making influence toward the new government-directed approval framework. Bill 40 also includes important transition and approval changes. A transitional provision exempts facilities that submitted a connection request before June 3, 2025 (the date the bill was introduced) so projects already in the queue are grandfathered. The Ontario Energy Board Act, 1998 was amended to add “economic growth” as a factor in leave-to-construct applications and the Municipal Franchises Act replaces municipal elector votes with council by-law approval for granting public utility rights. These changes streamline municipal approval for energy infrastructure. However, a critical gap exists given that regulations have not yet been published. Those regulations will define the approval criteria, community obligations and reporting requirements that govern specified load facilities. Bill 40 provides the framework; the regulations will determine how it operates in practice. Until they are finalized, developers may not know precisely what they must demonstrate to obtain a connection. The Data Centre Playbook: Three pillars for project evaluation On August 13, 2026, Ontario launched the Data Centre Playbook, a policy framework built on three pillars: (i) advancing economic development, (ii) protecting data sovereignty and (iii) investing in communities. The Playbook translates those pillars into a two-part evaluation for regulated data centres seeking grid connection: a System Impact Assessment (SIA), which examines connection feasibility, grid impacts and full cost recovery, including plans for on-site generation; and a Strategic Priority Assessment, which scores the project against the three pillars. Ontario’s stated objective is to “compete on value, not incentives.” The related 30-day public consultation through the Environmental Registry of Ontario (ERO Notice 026-0853) closed September 12, 2026. Under the economic-development pillar, Ontario will favour projects that create jobs, generate tax revenue, strengthen domestic supply chains and, preferably, are Canadian-owned and operated. Projects are also expected to make compute capacity available to Ontario businesses, including small and medium-sized enterprises. The government projects CA\$122 billion in economic growth by 2035 and more than 17,000 new jobs annually from AI-related development. At present, the Playbook emphasizes sovereignty as a policy consideration rather than imposing a general statutory data-localization requirement. Industry commentary has highlighted the privacy advantages of Canadian data centres and the broader case for digital and AI sovereignty. In the context of heightened Canada–US tensions, where data is stored, processed and controlled may be material to public-sector and regulated customers. Ontario has urged the federal Minister of Artificial Intelligence and Digital Innovation to prioritize digital sovereignty and strengthen protections for consumer data. Community investment is the third pillar. The Playbook contemplates significant financial and non-financial benefits for host communities, including (i) apprenticeship opportunities for mid-career workers; (ii) amenities such as community centres, parks and local roads; (iii) critical infrastructure upgrades, including broadband and grid improvements; and (iv) local workforce training and participation requirements. These commitments should be addressed during siting and project development, not treated as a post-approval add-on. Grid access will come with obligations. One of the most consequential proposals under consultation is a new Class C electricity rate for data centres above 1 MW, set higher than the existing industrial rate. The proposal is significant because it would make very large computational loads bear a distinct electricity cost rather than accessing industrial pricing by default. The government is considering amendments to Ontario Regulation 429/04 and excluding future large data centres from the Industrial Conservation Initiative (ICI), which allocates Global Adjustment costs among large industrial consumers. Ontario will offer only non-financial support, and data centres are expected to pay full electricity costs. Developers should model the rate, full cost recovery, locational pricing, connection costs and any ICI exclusion together. Power procurement will be a parallel workstream. The Playbook encourages on-site generation, and developers may also consider corporate power purchase agreements (PPAs) to secure long-term supply or hedge price exposure.

Neither approach removes the need to satisfy connection, technical and regulatory requirements. Environmental and social constraints will also shape siting and operations. Projects should assess water consumption, closed-loop cooling, heat-reuse potential, noise and carbon regulation, alongside environmental approvals, Indigenous engagement and the community's social licence. These issues can affect permitting, operating costs and the ability to secure local support. IESO technical requirements for large computational loads If Bill 40 determines who may connect and the Playbook sets the policy terms, the IESO's Technical Requirements determine how a large computational load must connect and operate. Version 3.0 applies to all computational loads rated above 10 MW connected to the transmission or distribution system, where the final System Impact Assessment or Notice of Conditional Approval was issued after June 30, 2026. "Computational load" means load from IT computing equipment, including servers, data storage and networking hardware, and covers data centres, AI compute clusters and cryptocurrency mining operations. These facilities behave differently from traditional industrial loads. IT equipment is voltage-sensitive and may switch to uninterruptible power supply systems during even brief voltage drops, producing sudden swings in grid demand. Rapid ramping, periodic oscillations, frequent fluctuations and abrupt reconnection after outages compound the reliability challenge. The IESO uses a two-stage interconnection process. Before equipment vendors are selected, it conducts steady-state studies and issues a comfort letter with a draft report identifying available capacity. After vendors are selected, a full System Impact Assessment and Notice of Conditional Approval follow. This sequencing lets developers test feasibility before committing capital, while reserving detailed compliance for the equipment-specific stage. Key requirements include: no single point of failure may cause the loss of more than 600 MW of load; power factor between 0.9 lagging and 0.9 leading; a maximum ramp rate of 20 MW per minute and no more than 5 MW of change in any 15-second period; continuous operation between 58.8 Hz and 61.2 Hz; and separate voltage ride-through requirements for transmission and distribution connections. Facilities above 25 MW must participate in automatic under-frequency load shedding, and projects of 20 MVA or greater must install phasor measurement units. Version 3.0's most important change is that it extends these requirements to distribution-connected projects, not just transmission-connected projects. The requirements include an enforcement mechanism under which the IESO may require corrective action and, in certain circumstances, direct disconnection through the applicable transmitter or distributor. Developers should treat technical design, equipment procurement and commissioning as part of the approval strategy from the outset. Stakeholder implications: Approvals, cost and project execution Developers now face three linked gates: (i) ministerial approval under Bill 40; (ii) IESO technical compliance; and (iii), if finalized, the Class C rate and full cost recovery with the OEB. With the exception of Bill 40, none of this is unique to load connections, however, these requirements are now tailored to a specific type of load and ultimately, political sign-off. The project case must address job creation, domestic supply-chain commitments, compute availability, community benefits, Canadian ownership and energy procurement, not just land and building economics. Depending on ownership structure and transaction characteristics, certain projects may also warrant early consideration of Investment Canada Act and national security review issues, particularly where a project involves critical infrastructure, sensitive data, national-security considerations or control by a state-linked investor. Developers should model higher electricity costs and possible locational price exposure. Behind-the-meter generation and corporate PPAs may provide supply certainty or a hedge, but neither substitutes for grid approvals. Ratepayers are protected by design: the full-cost-of-power principle, the proposed Class C rate and the potential Industrial Conservation Initiative exclusion are intended to prevent data centres from receiving subsidized electricity at other consumers' expense. Notably, the Playbook also provides that data centres should not hinder growth in other sectors, such as manufacturing. For more information about how these developments may affect your business, please contact the authors, Kristyn Annis and Solomon Ezike.

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